

## Product Feature

1. 15\_bit absolute encoder, pulse up to 32768。
2. Multi-stage DD motor structure, large torque output。
3. Integrated servo, simplified wiring, small size。
4. Low noise, low vibration, high speed positioning, high reliability.
5. FOC field orientation vector control, support position/speed closed loop.
6. Can work in zero lag given pulse state, follow zero lag.
7. 16 position electronic gear function.
8. modbus RTU /CANOPEN
9. Position mode, support pulse + direction signal
10. With blocked rotation,overcurrent protection, overvoltage protection.



## Parameter List

| Model Specification        |                               | 57AIM15                                                                    | 57AIM15H                                    | 57AIM30                                     | 57AIM30H                                    |
|----------------------------|-------------------------------|----------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| power supply               | voltage                       | 24~36VDC                                                                   | 24~36VDC                                    | 24~36VDC                                    | 24~36VDC                                    |
|                            | electricity                   | 2.2A                                                                       | 2.2A                                        | 4.4A                                        | 4.4A                                        |
| electric machine parameter | torque                        | 0.48NM                                                                     | 0.24NM                                      | 0.96NM                                      | 0.48NM                                      |
|                            | Rated speed                   | 1000RPM                                                                    | 2500RPM                                     | 1000RPM                                     | 2500RPM                                     |
|                            | Maximum speed                 | 1500RPM                                                                    | 3000RPM                                     | 1500RPM                                     | 3000RPM                                     |
|                            | power                         | 50W                                                                        | 50W                                         | 100W                                        | 100W                                        |
|                            | resistance                    | 2.65 Ω                                                                     | 2.65 Ω                                      | 1.3 Ω                                       | 1.3 Ω                                       |
|                            | inductance                    | 1.1mH                                                                      | 1.1mH                                       | 0.5mH                                       | 0.5mH                                       |
|                            | moment of inertia             | $9.139 \times 10^{-5}$<br>KG/M <sup>2</sup>                                | $9.139 \times 10^{-5}$<br>KG/M <sup>2</sup> | $1.184 \times 10^{-5}$<br>KG/M <sup>2</sup> | $1.184 \times 10^{-5}$<br>KG/M <sup>2</sup> |
|                            | feedback signal               | Multi-turn absolute encoder ( 32768 pulses per lap, 15 positions per lap ) |                                             |                                             |                                             |
| type of cooling            |                               | air cooling                                                                |                                             |                                             |                                             |
| weight                     |                               |                                                                            |                                             |                                             |                                             |
| Position control mode      | Maximum input pulse frequency | 500KHz                                                                     |                                             |                                             |                                             |
|                            | Pulse command mode            | Pulse + direction, phase A + phase B                                       |                                             |                                             |                                             |
|                            | Electronic gear ratio         | The value ranges from 1 to 65535                                           |                                             |                                             |                                             |

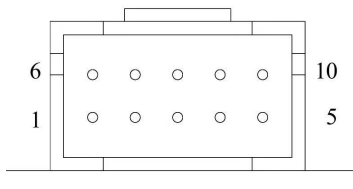
|                                |                                         |                              |
|--------------------------------|-----------------------------------------|------------------------------|
|                                | <b>Position sampling frequency</b>      | 2KHz                         |
| <b>protect function</b>        |                                         | Motor blocking alarm         |
| <b>communication interface</b> |                                         | RS485 (modbus RTU) / CANOPEN |
| <b>working environment</b>     | <b>environment temperature</b>          | 0~40°                        |
|                                | <b>Motor allows maximum temperature</b> | 85°                          |
|                                | <b>humidity</b>                         | 5~95%                        |

## Interface Function

port number: Facing the port, the left side is the first.

| Port number | name       | function                                                                                                                                                                                                                                                                   |
|-------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | +24V       | Dc power supply positive, +24V. The positive and negative connection will directly short circuit the power supply and may damage the driver                                                                                                                                |
| 2           | GND        | Direct current source. The positive and negative connection will directly short circuit the power supply and may damage the driver                                                                                                                                         |
| 3           | PU+ (+5V)  | Pulse control signal: pulse rising edge effective; PU- high power usually 3.3 ~ 5V, low power usually 0 ~ 0.5V.<br>In order to reliably respond to the pulse signal, the pulse width should be greater than 1.2μs. If +12V or +24V is used, a series resistor is required. |
| 4           | PU- (PU)   |                                                                                                                                                                                                                                                                            |
| 5           | DIR+ (+5V) | Direction signal: high/low level signal, in order to ensure the reliable commutation of the motor, the direction signal should precede the pulse signal<br><br>Establish at least 5μs. DIR- 3.3 ~ 5V in high power, 0 ~ 0.5V in low power.                                 |
| 6           | DIR- (DIR) |                                                                                                                                                                                                                                                                            |

Port number: Facing the port, the lower row is 12345 from left to right, and the upper row is 6 7 8 9 10 from left to right.



The 485 version plug is defined as follows:

| Port number | name | function            |
|-------------|------|---------------------|
| 1           | NC   |                     |
| 2           | 485A | 485 communication + |
| 3           | 485B | 485 communication - |
| 4           | EN+  | EN+                 |

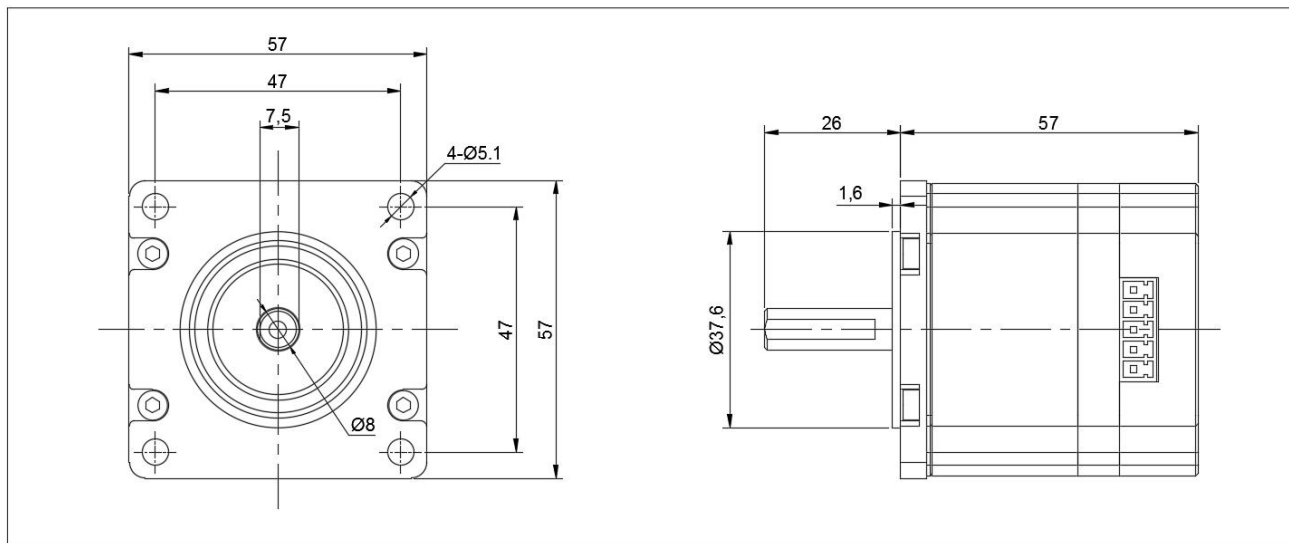
|    |        |                                                                                                                                                                                                                                                                            |
|----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5  | EN-    | EN-                                                                                                                                                                                                                                                                        |
| 6  | COM    | The output signal is grounded with the 485 power supply.                                                                                                                                                                                                                   |
| 7  | WR     | Alarm signal output, the internal optical coupling NPN output. Normally, it is in high resistance state, and COM is on with alarm.                                                                                                                                         |
| 8  | RDY/PF | Servo ready signal/position signal. There is a signal (conduction) after the power-on automatic work, when the following error is less than 0.5°, there is a signal (conduction), and when the following error is greater than 0.5°, there is no signal (high resistance). |
| 9  | ZO     | Encoder zero output. There is zero signal optical coupling NPN output guide communication number                                                                                                                                                                           |
| 10 | 485_5V | 485 Communication 5V power supply. External power supply is required. (This power supply is supplied through the controller)                                                                                                                                               |

CAN version plug is defined as follows:

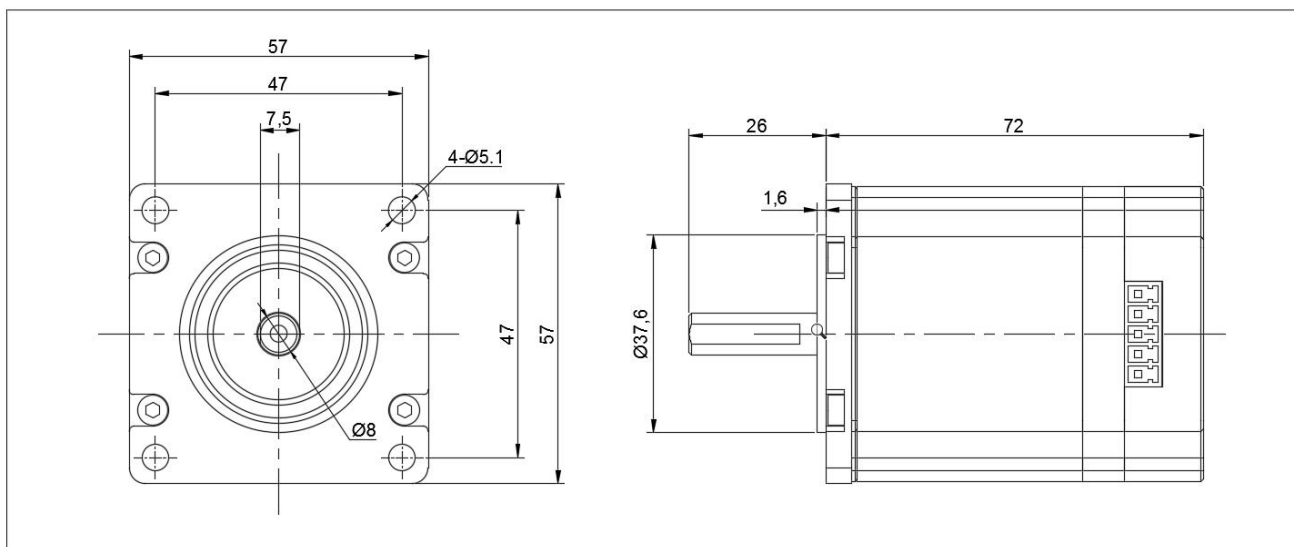
| Port number | name   | function                                                                                                                                                                                                                                                                   |
|-------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | CANL   | Can communication port, Use CAN communication, CAN_5V,COM need 5V                                                                                                                                                                                                          |
| 2           | NC     |                                                                                                                                                                                                                                                                            |
| 3           | NC     |                                                                                                                                                                                                                                                                            |
| 4           | CANH   | Can communication port, Use CAN communication, CAN_5V,COM need 5V                                                                                                                                                                                                          |
| 5           | NC     |                                                                                                                                                                                                                                                                            |
| 6           | COM    | Output signal and CAN Ground connection                                                                                                                                                                                                                                    |
| 7           | WR     | Alarm signal output, the internal optical coupling NPN output. Normally, it is in high resistance state, and COM is on with alarm.                                                                                                                                         |
| 8           | RDY/PF | Servo ready signal/position signal. There is a signal (conduction) after the power-on automatic work, when the following error is less than 0.5°, there is a signal (conduction), and when the following error is greater than 0.5°, there is no signal (high resistance). |
| 9           | ZO     | Encoder zero output. There is zero signal optical coupling NPN output guide communication number.                                                                                                                                                                          |
| 10          | CAN_5V | CAN communication 5V power supply, external power supply is required. (This power supply is supplied through the controller)                                                                                                                                               |

## Motor Size

57AIM15:



57AIM30:



57AM30B (刹车)

